

Lap 1

1. What is the difference between less and more?

- more → mainly moves forward through the file.
- less → gives you more control, including moving backward and searching.

2. What is the difference between cat and more command?

The main difference is how the file content is displayed:

- cat --> Displays the entire file at once
- more --> Displays the file page by page

3. What is the difference between rm and rmdir using man?

- rm → remove files (and directories with options)
- rmdir → remove directory, but only if empty

4. Create the following hierarchy under your home directory:

```
hajer@hagar-Victus:~$ cd dir1/
hajer@hagar-Victus:~/dir1$ mkdir dir11
hajer@hagar-Victus:~/dir1$ mkdir dir12
hajer@hagar-Victus:~/dir1$ cd
hajer@hagar-Victus:~$ mkdir docs
hajer@hagar-Victus:~$ cd docs/
hajer@hagar-Victus:~/docs$ mkdir mycv
hajer@hagar-Victus:~/docs$ cd
hajer@hagar-Victus:~$ cd dir1/dir11
hajer@hagar-Victus:~/dir1/dir11$ touch file1.txt
hajer@hagar-Victus:~/dir1/dir11$ ls
file1.txt
hajer@hagar-Victus:~/dir1/dir11$ cd ..
hajer@hagar-Victus:~/dir1$ tree
.
├── dir11
│   └── file1.txt
└── dir12

3 directories, 1 file
hajer@hagar-Victus:~/dir1$
```

a. Remove dir11 in one-step. What did you notice? And how did you overcome that?

```
hajer@hagar-Victus:~$ rmdir dir1/dir11
rmdir: failed to remove 'dir1/dir11': Directory not empty
hajer@hagar-Victus:~$ rm -r dir1/dir11
hajer@hagar-Victus:~$ cd dir1/
hajer@hagar-Victus:~/dir1$ ls
dir12
hajer@hagar-Victus:~/dir1$
```

- rmdir can remove only empty directories.
- use rm -r because -r means recursive, so it removes the directory and everything inside it.

b. Then remove dir12 using rmdir -p command. State what happened to the hierarchy (Note: you are in your home d

```
hajer@hagar-Victus:~/dir1$ cd
hajer@hagar-Victus:~$ rmdir -p dir1/dir12
hajer@hagar-Victus:~$ ls
Desktop  Documents  Downloads  Music  Nti  Pictures  Projects_  Public  Templates  Videos  docs  snap
hajer@hagar-Victus:~$
```

rmdir -p dir1/dir12 removes dir12 and then removes its empty parent directory

c. The output of the command pwd was /home/user. Write the absolute

and relative path for the file mycv

Absolute path: /home/user/docs/mypcv Relative path: docs/mypcv

5. Copy the /etc/passwd file to your home directory making its name is mypasswd.

```
hajer@hagar-Victus:~$ ls
Desktop  Documents  Downloads  Music  Nti  Pictures  Projects_  Public  Templates  Videos  docs  snap
hajer@hagar-Victus:~$ pwd
/home/hajer
hajer@hagar-Victus:~$ cp /etc/passwd ~/mypasswd
hajer@hagar-Victus:~$ ls
Desktop  Documents  Downloads  Music  Nti  Pictures  Projects_  Public  Templates  Videos  docs  mypasswd  snap
hajer@hagar-Victus:~$ ls -l mypasswd
-rw-r--r-- 1 hajer hajer 2921 Aug 27 15:35 mypasswd
hajer@hagar-Victus:~$
```

6. Rename this new file to be oldpasswd.

```
hajer@hagar-Victus:~$ mv mypasswd oldpasswd
hajer@hagar-Victus:~$ ls
Desktop  Documents  Downloads  Music  Nti  Pictures  Projects_  Public  Templates  Videos  docs  oldpasswd  snap
hajer@hagar-Victus:~$
```

7. You are in /usr/bin, list four ways to go to your home directory

```
hajer@hagar-Victus:~$ cd /usr/bin/
hajer@hagar-Victus:/usr/bin$ cd
hajer@hagar-Victus:~$ cd /usr/bin/
hajer@hagar-Victus:/usr/bin$ cd ~
hajer@hagar-Victus:~$ cd /usr/bin/
hajer@hagar-Victus:/usr/bin$ cd $HOME
hajer@hagar-Victus:~$ cd /usr/bin/
hajer@hagar-Victus:/usr/bin$ cd /home/hajer/
hajer@hagar-Victus:~$
```

8. List Linux commands in /usr/bin that start with letter w

```
hajer@hagar-Victus:/usr/bin$ cd /home/hajer/
hajer@hagar-Victus:~$ cd /usr/bin/
hajer@hagar-Victus:/usr/bin$ ls w*
w          watchgnupg  wdctl      whereis    who         wireplumber  wpexec
waitpid    watchmedo  websockets which       whoami      word-list-compress  wsdd
wall       wc         wget       which.debianutils  whoopsie    wpa_passphrase
watch      wcurl     whatis     whiptail    whoopsie-preferences  wpctl
```

9. Display the first 4 lines of /etc/passwd

```
hajer@hagar-Victus:~$ head -n 4 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
hajer@hagar-Victus:~$
```

10. Display the last 7 lines of /etc/passwd

```
hajer@hagar-Victus:~$ tail -n 7 /etc/passwd
rtkit:x:978:978:RealtimeKit:/proc:/usr/sbin/nologin
colord:x:977:977:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
gdm:x:975:975:Gnome Display Manager:/var/lib/gdm3:/bin/false
hajer:x:1001:1001:hajer:/home/hajer:/bin/bash
hajer_test:x:1002:1002::/home/hajer_test:/bin/sh
user100:x:1000:1000:,,,:/home/user100:/bin/bash
sshd:x:973:65534:sshd _user:/run/sshd:/usr/sbin/nologin
```